

Chapter 2

Epidemiological Studies and What They Really Mean

Definitive experiments are rare in the fields of science and medicine. Instead, conclusions generally emerge from the integration and synthesis of evidence from a variety of sources. Using a similar approach, lawyers argue cases in which each piece of evidence is considered circumstantial in the hope that eventually the circumstantial evidence will become overwhelming. Like lawyers, scientific researchers gather and regather circumstantial evidence from several perspectives with the goal of understanding cause and effect. By studying the relationships among variables and investigating mechanisms, they are able to form perspectives that are robust and plausible. This type of research work, together with longitudinal studies, tests the causative factors identified in the mechanistic studies.

Other chapters in this book address the mechanisms of the low back and their links to good health and disability. This chapter focuses on the study of associations of variables through various epidemiological approaches. Some readers will find this a boring chapter—so my students inform me. You may choose to skip this chapter. But before you do, consider that I will show you that many of these studies have been misinterpreted, or misrepresented, to serve the agenda of a group. Those wishing a fuller understanding of the challenges of building a strong foundation for the very best injury prevention and rehabilitation programs are encouraged to read on. Doing so will enable you to appreciate the epidemiological perspective, to have a more complete comprehension of the positions taken in the text as a whole to reduce the economic impact of low back disorders, and to understand certain subsequent approaches for prevention and rehabilitation.

Several sections in this chapter will help you understand the risk factors for low back trouble—specifically, the changes in personal factors and whether they cause back troubles or are a consequence of having them. For the purpose of this review, purported disabling low back troubles and possibly related work intolerance will be referred to collectively as low back disorders (LBDs). Furthermore, the term personal factors can include

anthropometric and fitness variables, as well as motor control ability, injury history, and so forth.

Influence of Randomized Controlled Trials and Other Epidemiological Approaches

What is the truth regarding back pain cause and treatment? Generally, the randomized controlled trial (RCT) is regarded as the gold standard in medicine. However, this has led to several questionable popular opinions regarding back pain. This section addresses the role of RCTs for back pain.

Back pain is not a homogeneous condition. Any therapeutic approach that relieves one person's pain will exacerbate another's. Epidemiological studies of back pain will never reveal cause and effect, or efficacy, because each person will respond to a different approach and different dosage. Thus, a controlled study on back pain will result in the conclusion of no effect. However, when patients are categorized into subgroups based on pathomechanics, or pain patterns, or history, or even psychosocial variables, results in the evolving literature are generally positive. Fersum, Dankaerts, and O'Sullivan (2009) opined that "non-specific exposures, with non-specific treatments, with non-specific back pain guarantees a null finding." Each patient is unique. Patients sit differently and use different strategies that either keep them in pain or out of pain (Dankaerts et al., 2006), and they react differently to different movement corrections (Ikeda and McGill, 2012). By their very nature, RCTs are not suitable for unclassified low back pain.

The evolution of the spine literature lags behind that on the knee and related leg pain. As an exercise in logic, let's replace the discussion of back pain with that of leg pain and observe how the back literature has been misinterpreted. No study of nonspecific leg pain would be expected to reveal cause and effect (the genesis of pain could be due to mechanical, vascular, hereditary, or many other factors). Similarly, no credible person would use such logic to state that there is no evidence of links between mechanical factors and nonspecific leg pain. Nor would such an assertion be published. However, a very specific leg disorder such as ACL disruption in female basketball players has been substantially reduced by a specific intervention to alter biomechanics (e.g., Hewett et al., 1999). One could argue that the treatment effect is underestimated. Even in these controlled clinical trials, everyone in the group received the same treatment. However, in excellent clinical practice, each intervention is tuned and adjusted to each patient, suggesting that efficacy in real life on a patient-by-patient basis is probably

better than any controlled trial would suggest. Well-presented examples of the interactions among individual anatomical, morphological, mechanical, and neuromechanical factors have been compiled by McLean and Beaulieu (2010), implying the need for individual assessment and treatment.

Some have opined that the relationships studied for the leg do not exist for the spine. Yet evidence exists on disc shape, to choose just one example from our own laboratory (Yates, Giangregorio, and McGill 2010), influencing the patterns of annulus disruption from specific modes of loading resulting from specific movement patterns. This supports individual assessment of anatomical features to justify appropriate clinical treatment to alter offending movement. The complex interactions of structure and biomechanics suggest that evidence from controlled trials of back pain must be coupled and interpreted with treatment studies of appropriately subclassified patients, even to the point of case studies, to delineate the complex interactions. I hope this convinces you that the back pain literature needs a higher level of perspective to converge on the truth.

The mechanism of injury must be considered before a systematic review on back pain is valid. For example, discogenic back injury and pain are associated with repeated and prolonged disc flexion. This occurs in occupational lifting from lower heights, but it also occurs in office workers who sit for prolonged periods. Both scenarios are potent causative factors (see a review by Adams et al., 2002). Reviewing studies that compared workers engaged in lifting activities to those who do not lift but have sedentary jobs (i.e., sitting) does not test lifting and pain, but instead tests the mechanism (i.e., disc flexion). Many reviews ignore known mechanisms that obscure any association or cause and effect. A textbook such as Marras' *The Working Back* (2008) integrates the mechanisms and defines links between back disorders and occupational exposure—for example, the hazards of lifting from the floor versus from table height, particularly for aging workers with stiffer hips. These concerns have motivated others to call for a more extensive study of the mechanisms of back pain to provide a better base for what is tested in an RCT (Gordon, 2010). The dose is important, but what is the effective dose?

Another widely held clinical belief is that a diagnostic test must be reliable to be of value (for a full essay on the topic, see McGill, 2013). This means that five clinicians would reach the same conclusion testing the same patient. Simple tests are reliable. Measuring core temperature in a case of

infection, or blood chemistry in a diabetic, are reliable tests that lead to a homogeneous treatment. But musculoskeletal disorders are different: the subjects are highly variable in terms of pain, there is often more than one source of pain, the dosage of intervention is critical (because too much exacerbates and too little has no effect), and the outcome is highly variable in terms of duration and effectiveness. Why should two clinicians obtain the same impression when examining a biological system that is continually in a state of flux? If skill was equal, and the patient remained static, then prediction may be possible and even justifiable. But this is not the case with most musculoskeletal disorders. Thus, the typical rules for reliability associated with evidence-based medicine need a liberal amount of reflection for logical application in back pain situations.

Do clinicians have similar levels of clinical skill? Clinical skill involves perceiving via touch; interpreting what the patient verbalizes and displays with body language; knowing how much force to apply; and knowing how to explore the end range and arc of motion with subtle trajectory variations to interpret joint capsule, bony interaction, ligament spring, and associated muscle tone, to name just a few variables. A simple test, such as one to objectify the range of motion, becomes much more revealing in the hands of a skilled clinician who probes tissues through a range. A less skilled clinician will overlook modulators of pain and function and should not be expected to replicate the clinical impression.

So what is the better test? The simple test to obtain a range of motion score that is reliable, or the test that facilitates a branching decision tree using variables and relationships that are nonlinear, change over time, and are not repeatable among clinicians of differing skill levels? Should reliability be used to identify a good test, and by logical extension, should reliability be a metric for inclusion into clinical guidelines? This argument illustrates how skilled diagnosticians using less reliable tests but more complex decision trees will obtain more insight into the patient than unskilled ones.

Tragically, the more complex cases, many of which fail to be sorted out by clinical guidelines advocating reliable diagnostic tests, are dismissed by the health care system. I see many of these sad cases in my consulting practice. Skilled clinicians must be encouraged to maintain and develop their skills in the more complex, but more insightful (but unreliable) assessment tests. Over my 30 years of observation, my impression is that this has declined. Unfortunately, political forces via medical journal publication

policies are influencing the development of clinical skill. I am against the growing emphasis on developing only tests that are reliable, and I am even more against the practice of journals now requiring a high test reliability score to qualify for publication. This practice is retarding expert assessment skill acquisition and development, and optimal patient outcomes for all branches of manual medicine.

Misunderstandings of Epidemiology

Epidemiological studies have been used almost exclusively to drive policy change that affects back pain issues. The compensation board in New Zealand (Accident Compensation Corporation) now denies any association between the type of work exposure and the development of disabling back pain. How could this be, given the multitude of studies confirming the link, and opposing policies of other countries? They appear to have based their position and policy on a study of twins (Battie et al., 2009), claiming that genetics explains the majority of variance. This study assessed MRI images of 147 monozygotic twins and 153 dizygotic twins that had exposure discordance. It was estimated that 61% of variance in disc degeneration could be explained by their common genes, whereas 16% was explained by occupation. McLean and colleagues (2011) stated that this was based on a misunderstanding. They noted that variation is not the same as causation, and that what is inherited is not the same as genetic determination. They give examples of specific, well-documented, inherited diseases that occurs only in the presence of the specific exposure trigger. Thus, the cause is both 100% genetic and 100% environmental. They also used smoking to illustrate the complexities of ferreting out exposure. If every person smoked one pack of cigarettes per day, smoking would not emerge as a causative factor. Only genetic predisposition to smoke would appear significant. Thus, the finding that 61% of variation in disc degeneration is explained by genetics does not mean that 61% of back pain is caused by genetics; in fact, it does not show anything in terms of causation. The same principle holds true for excessive noise exposure and hearing loss, and exposure to radiation and cancer.

As my good colleague Professor Marras has stated, "Genetics loads the gun, but exposure pulls the trigger." However, there is a difference with back pain. Exposure to physical loads is not a linear relationship—no load and too much load are both harmful. Some load is healthful. This will become clear in the following pages.

Multidimensional Links Among Biomechanical, Psychosocial, and Personal Variables

As noted in chapter 1, several prominent people have declared that psychosocial variables are the most significant factors in LBD. This is an important issue because, to be effective, intervention must address the real causes and consequences of back troubles. This section shows that virtually all studies that properly measured or calculated the physical demands of tasks revealed that people subjected to specific mechanical stressors are at higher risk of LBD than others, but that there also appear to be some mitigating issues.

Four Important Studies

The following four studies are reviewed here because of their lasting historical influence.

- Bigos and colleagues. In 1986 Bigos and colleagues performed a highly quoted study—one that has been very influential in shaping opinion regarding injury prevention and rehabilitation—at the Boeing plant in Washington state in the United States. This retrospective investigation analyzed 4,645 injuries (of which 900 were to the low back) over a 15-month period in 1979-1980. The authors reported a correlation between the incidence of back injury and poor appraisal ratings of employees performed by their supervisors within 6 months prior to the reported injury. The authors considered the poor ratings to represent a psychosocial factor. In 1991 Bigos and colleagues conducted a longitudinal prospective study of 3,020 employees at Boeing, during which there were 279 reported low back injuries. The researchers collected personality inventories as well as questionnaires regarding family and coworker support and job satisfaction. They also analyzed personal factors such as isometric strength, flexibility, aerobic capacity, height, and weight. The authors concluded that psychosocial measures—particularly those related to job enjoyment—had the strongest influence of all the variables analyzed. In fact, those workers who stated that they

did not enjoy their job were 1.85 times more likely to report a back injury (odds ratio = 1.85). Job satisfaction counted for less than 15% of the variance as an injury risk factor, meaning that more than 85% of the variance was unaccounted for. In other words, psychosocial factors failed to account for 85% of the causation of LBDs. The authors concluded “that the statistically significant, though clinically modest, predictive power of work perceptions and psychosocial factors for reports of acute back pain among industrial workers argues against the exclusive use of an injury model to explain such problems.” This is a fair summary of the implications of their work. It does not mean that mechanical loading is unimportant. Yet, this study is often quoted to support the viewpoint that psychosocial factors are the most important causes of back disorders. Interestingly, Marras and colleagues (1993) also found similar odds ratios for job satisfaction (1.56) in a massive study of 400 repetitive industrial lifting jobs across 48 industries.

Very few epidemiologically based studies have employed reasonably robust quantifications of biomechanical, psychosocial, and personal factors. Two important studies meet this requirement.

Odds Ratios

Perhaps the most lucid definition of an odds ratio can be achieved through an example. If smokers have three times the risk of developing lung cancer that nonsmokers do (perhaps 6 out of 10 smokers as opposed to 2 out of 10 in nonsmokers), they have an odds ratio of 3. Thus, an odds ratio greater than 1 suggests an increased risk from a specific factor.

- Marras and colleagues. The first important study **that considered multiple factors** was reported by Marras and colleagues (1995), who surveyed over 400 industrial lifting jobs across 48 industries. They examined medical records in these industries to classify each type of job as being low, medium, or high risk for causing LBD. They documented a variety of mechanical variables as well as reporting job satisfaction. The most powerful variable for predicting those jobs with LBD was maximal low back moment. This resulted in a predictive odds ratio of

4.04 between low- and medium-risk groups and a ratio of 3.32 between the low- and high-risk groups. Other single variables produced impressive odds ratios; for example, sagittal trunk velocity (odds ratio = 2.48) for the low- and high-risk comparison and 2.42 for maximal weight handled between the low- and high-risk groups. Job satisfaction produced an odds ratio of 1.48 between the low- and high-risk groups and of 1.32 between the low- and medium-risk groups. The researchers entered the single variables into a multiple logistic regression model. The group of measures selected by the model described the risk index well between the low- and high-risk groups and also between the low- and medium-risk groups. Suitably varying the five measures chosen by the regression process (maximal load moment, maximal lateral trunk angular velocity, average trunk twisting velocity, lifting frequency, and maximal sagittal trunk angle) decreased the odds of being a member of the high-risk LBD group over 10 times (odds ratio = 10.6). This was an important study for specifying certain physical characteristics of job design that reduce the risk of LBD and linking epidemiological findings with quantitative biomechanical analysis and psychosocial factors across a large working population. Marras' group has continued with similar studies to strengthen the understanding of biomechanical and psychosocial risk factors in predicting meaningful low back functional losses (e.g., Ferguson et al., 2012).

- Norman and colleagues. The second important study to successfully integrate biomechanical, psychosocial, and personal factors was conducted by Norman and colleagues (1998), who examined injuries that occurred in an auto assembly plant that employed more than 10,000 hourly paid workers. During a 2-year period of observation in the plant, the authors reported analyses on 104 cases and 130 randomly selected controls. Cases were people who reported low back pain (LBP) to a nursing station; controls were people randomly selected from company rosters who did not report pain. This is a notable study because the authors attempted to obtain good-quality psychosocial, personal, and psychophysical data on all participants from an interviewer-assisted questionnaire, as well as good-quality, directly measured biomechanical data on the physical demands of the jobs of all participants. (Note that the psychophysical approach is based on worker self-perceived stresses.) The study revealed that several independent and highly significant

biomechanical, psychosocial, and psychophysical factors (identified as risk factors) existed in those who reported LBP. The personal risk factors that were included were much less important. After adjusting for personal risk factors, the statistically independent biomechanical risk factors that emerged were peak lumbar shear force (conservatively estimated odds ratio = 1.7), lumbar disc compression integrated over the work shift (odds ratio = 2.0), and peak force on the hands (odds ratio = 1.9). The odds ratios for the independent psychosocial risk factors, from among many studied, were worker perceptions of poorer workplace social environment (2.6), higher job satisfaction (not lower, as shown in the Boeing study) (1.7), higher coworker support (1.6), and perception of being more highly educated (2.2). Perceptions of higher physical exertion, a psychophysical factor, resulted in an odds ratio of 3.0, which is possibly related to the capacity of the worker relative to the job demands. Nearly 45% of the total variance was accounted for by these risk factors, with approximately 12% accounted for by the psychosocial factors and 31% by the biomechanical factors. These results are very consistent with those reported by Marras and colleagues (1995) and by Punnett and colleagues (1991). Only a few of the personal factors were associated with reporting LBP: body mass index (odds ratio = 2.0) and prior compensation claim (odds ratio = 2.2). This case-control study is of high quality because it used a battery of many of the best measurement methods available for field use to assess many psychosocial, biomechanical, and personal factors of all participants in the data pool.

- Coenen and colleagues. This Dutch group (2013) studied the spine loads of workers and documented personal and psychosocial factors. Three years later, a follow-up documented the occurrence of low back pain. Cumulative low back load had a greater association with the onset of pain than any other factor including those considered as personal and psychosocial. Interestingly, a more detailed analysis revealed that this relationship was strengthened when workers spent more time with their spines flexed and lifts of loads weighing more than 25 kg (55 lb). The authors' position on the importance of motions, postures, and loads causing low back pain strengthens throughout this monograph.

The evidence from the preceding comprehensive studies suggests that

both psychosocial and biomechanical variables are important risk factors for LBD. In particular, cumulative loading, joint moments, and spine shear forces are important. Those claiming that only psychosocial factors are important or that only physical loading factors are important cannot mount a creditable, data-based defense, because it appears that the data they quote fail to measure properly either physical or psychosocial variables, or both.

Do Workers Experience LBDs Because They Are Paid to Act Disabled?

Some papers in the literature appear to dismiss the link between pain and disability. Most of these papers clearly state that this notion is restricted to nonspecific back pain, noting that specific diagnoses do impair the ability of a worker to perform a demanding job. However, some authors base their arguments on the concept that low back tissue injury heals in 6 to 12 weeks, whereas others base their arguments on a behavioral model of chronic pain that is not totally consistent with the findings of other scientific approaches. A short discussion of the issues and evidence related to physician diagnosis, compensation, tissue damage, and pain is necessary.

The position that chronic pain and disability are a function of compensation (and not mechanical factors) is contradicted by evidence that low back troubles continue after legal settlement of injury compensation (Mendelson, 1982). Hadler, Tait, and Chibnall (2007) believe that the contest between the patient and the medical officer charged with determining the compensatory award causes the patient to act disabled and thwarts any incentive to act well. Teasell and Shapiro (1998) shared this opinion in a review of several chronic pain disorders, as did Rainville and colleagues (1997) in another high quality study that specifically addressed chronic LBP. Hadler has taken the position that mechanical factors are, for the most part, of little importance in either causing or rehabilitating bad backs when compared with the psychosocial modulators (e.g., Hadler, 2001). His selective citation of the literature excludes evidence linking mechanical overload to tissue damage and ignores several important intervention studies. For example, Werneke and Hart (2001) showed that pain patterns on patient presentation, specifically whether the pain “centralizes” or not, are much more powerful predictors of chronicity than the psychosocial variables they studied.

This rather common perception that injury compensation negatively affects the health recovery of injured workers has been challenged by Spearing and Connelly (2011). They reviewed the evidence often cited to support this perception for both verifiable and nonverifiable injury. They refuted the notion that strong evidence exists of an association between compensation and poorer health outcome, even for one of the most documented disorders—whiplash. They showed that the laws and legal processes are heterogeneous and that any link of a disorder to compensation is extremely difficult to control. Of the patients I have seen for whom compensation has been suggested to modulate their behavior and recovery, I have found an alternate explanation in nearly every case. In some, the rehab program assigned to them was ill-conceived and hurt them more. Some were never shown the mechanical causes of their pain. The jobs of others were so physically demanding that few healthy people would survive the daily rigors. These jobs demanded close to 100% of their healthy capabilities, meaning that 90% capability was not sufficient. Their doctors did not have the tools to measure the jobs' physical demands nor their capabilities. Yet some jurisdictions have started to move to limiting access to compensation, believing the association between health and compensation exists, is common, and is simple. It is not.

Of the several parties involved in the compensation system, all wish for a healthy patient. However, several factors conspire to militate against an optimal process and experience for all. There is much heartache and unfairness. Some unfortunate patients are rejected from the comp system because they fail to get better, or actually get worse, and are labeled noncompliant. This is largely because the comp system usually employs a rehab approach that brings finality to the case. This approach is often known as work hardening and is characterized by physical tasks that have a systematic schedule for increased challenge. Some claim that the type of challenge or exercise or work task is not important—only that it be performed. Patients are encouraged to work through the pain. There is no question that many patients thrive under this approach and are successfully discharged as employed workers. However, this approach is not for every patient. Typically, these sorts of programs have significant dropout rates. The comp system usually labels these people as noncompliant, and they are dismissed from payouts. Psychosocial issues are usually given as reason for their inability to cope. My opinion is different. Many of these people have

backs that may be unstable, and the work-hardening program incorporates the injury mechanism as part of the treatment and makes them worse. Butler (e.g., 1991, 2000) has documented for years that central sensitization mechanisms and secondary hyperalgesia are based on measurable changes in nervous structures and that more pain during movement only heightens the syndrome. The mechanisms of sensitization are nicely described by Woolf (2010). Because corrective exercise has not been prescribed, the tissues are further damaged, or at least prevented from healing. For these patients we work to eliminate the cause of their pain and ensure pain-free therapeutic exercises specifically designed to address their deficits and the actual cause of tissue overload. Even with people who have the most disabled of backs, those who have been classified as failures and labeled with no hope for recovery (i.e., 0% chance of returning to work), this approach gets about 35% back to work.

Thus, although the topic of compensation is important, it is irrelevant in discussions of the links between loading and LBD. Compensation issues should not be used to argue against the existence of a mechanical link between injury and work tolerance or, worse yet, to suggest that the removal of compensation will eliminate the cause of tissue damage.

Does Pain Have an Organic Basis—Or Is It All in the Head?

Much has been written about the apparent absence of an organic basis for chronic low back disability (and other chronic pain syndromes). As noted in chapter 1, as high as 85% of disabling LBD cases are claimed to have no definitive pathoanatomical diagnosis (White and Gordon, 1982). Two conclusions have been proposed:

- Many LBD patients present with “nonorganic signs,” suggesting psychological disturbance as the cause of their condition.
- Poor diagnostic techniques, either from inadequately trained doctors or from the limitations of widely available diagnostic technology, have precluded the making of many solid diagnoses.

The first conclusion suggests that the correct course of action in these

undiagnosed cases is to ignore physiological issues and address only psychological or psychosocial factors. The second conclusion suggests that one should not ignore the possibilities that more thorough diagnostic techniques could unearth physical causes and that rehabilitation based on that assumption could be more effective than rehabilitation for psychological disturbance alone. Let's examine each of these arguments.

Does Absence of Diagnosis Imply Psychological Cause?

Waddell and colleagues wrote many manuscripts and guidelines (e.g., 1980, 1984, 1987) on nonorganic signs in patients to support the notion of psychological disturbance overriding any pathoanatomical tissue damage (because none was diagnosed). Yet many of these so-called nonorganic signs are precisely what we will use clinically to detect hypersensitivity to loads that cause pain. For example, a compressive injury often creates pain under very mild compression when the person is in a flexed or slouched posture. Light touch and reported pain in a painful region may indicate a highly centrally sensitized patient. Although the nonorganic signs may be very helpful in defining risk for surgical candidates, we take the position that they do not indicate psychological overlay—in fact, the behavior may be physiologically based. Only a rigorous and complete exam that includes provocative testing could provide evidence to support the diagnostic hypothesis.

Teasell and Shapiro (1998) wrote a nice summary of an extensive experimental literature suggesting that these pain symptoms may indeed have a physiological basis. They reviewed the recent science on the spread of neuron excitability and sensitization of adjacent neurons to explain the sensation of radiating pain in chronic conditions. Changes in neuroanatomy are coupled with biochemical changes with chronic pain. For example, in fibromyalgia patients, numerous studies have shown levels of substance P in the cerebrospinal fluid elevated two to three times over that in controls (reported in a review by Teasell, 1997). Although the nonorganic signs described in Waddell's papers are important considerations in many cases and are a contribution to clinical practice, strong evidence suggests that many nonorganic signs may not be exclusive of a pathoanatomical mechanism that has eluded diagnosis.

Some have argued that no link exists between pain and tissue damage and

activity intolerance. In the absence of no direct evidence, some groups have simply assumed that nociceptive pain that is not surgically correctable or that has not improved within six weeks should not be regarded as disabling (e.g., Fordyce, 1995). In fact, the Fordyce monograph initiated much discussion, including several letters (e.g., Thompson/Merskey/Teasell/Fordyce, 1996). Fordyce's 1996 influential statements that "the course we are presently on threatens disaster" and that "we change or go broke" were particularly revealing. The high cost of treating chronic pain and disability appears to have motivated the elevation of the importance of the psychosocial factors so emphasized in this monograph. It is also interesting to observe the absence of any eminent biomechanical expert among the author list of the Fordyce report. The Canadian Pain Society stated that the Fordyce report literature review "is incomplete and does not reflect the contemporary understanding of chronic low back pain" (Thompson/Merskey/Teasell/Fordyce, 1996). Moreover, the Fordyce report largely ignored the evidence linking mechanical overload to measurable changes in spine biomechanics and spinal pain neuromechanical mechanisms. Another assumption of the report was that most spine LBDs are nonspecific, meaning simply that a diagnosis was not made.

Could Inadequate Diagnosis Be a Factor in Nonorganic LBP?

Bogduk and colleagues (1996) argued that pain arising from many spinal tissues can be attributed to a detectable painful lesion. For example, the facet joints produce pain upon stimulation (McCall, Park, and O'Brien, 1979). Bogduk's point is certainly correct. However, because not all lesions are easily detectable, one cannot argue that if a lesion is not detected, there is no organic basis for pain. For example, fractures and meniscal tears that have been detected in postmortem studies have not shown radiologically on planar X-ray (Jonsson et al., 1991; Taylor, Twomey, and Corker, 1990) or on computed tomography (CT) (Schwarzer et al., 1995). Nor have freshly produced fractures and articular damage been outwardly detectable radiographically in animal models (Yingling and McGill, 2000). Yet these are the typical diagnostic procedures used. Roudsari and Jarvik (2010) questioned the use of the MRI imaging procedure after reviewing the high prevalence of abnormal MRI findings in people with low back pain. However, the problem is more with the common practice of a clinician, most

often a surgeon, putting the images on the viewer before a thorough examination of the patient has been conducted. Of course, all sorts of features can be stated as the cause of the patient's symptoms, but this is poor practice. Better practice is to assess the patient and then hypothesize the pain source from the observed patterns. This diagnostic hypothesis is then refuted or confirmed with the MRI image.

Developments in technology such as fMRI have provided a deeper understanding of tissue irritation and pain sensitivity. Kobayashi and colleagues (2009) mechanically stimulated pain with controlled cutaneous pressure lateral to L4/L5 in both patients with chronic back pain and pain-free controls. The patients reported higher levels of pain, and also demonstrated higher levels of activation, in the prefrontal, insular, and posterior cingulate cortices but not in the S1 and S2 surface somatosensory areas, suggesting stimulation of painful deeper structures. These areas of brain activity are associated with the commonly reported pain matrix from a variety of truly painful nociceptive agents (Peyron, Laurent, and Garcia-Larrea, 2000). Back pain was a real organic phenomenon, but the elevated sensitization appears to be a trait of those in chronic pain. Later in this text, cleverly designed exercises are shown to be a potent approach for winding down the sensitivity.

Interestingly, Bogduk and colleagues (1996) and Lord and colleagues (1996) showed that the injection of an anesthetic (placebo-controlled diagnostic blocks) convincingly demonstrates that facet joints are often the site of pain origin. Further, disc studies examining the pain response of well over 1,000 discs in over 400 people undergoing discography by Vanharanta and colleagues (1987) (subsequently reappraised by Moneta and colleagues in 1994 and reported by Bogduk and colleagues in 1996) showed a clear and statistically significant correlation between disc pain and grade 3 fissures of the annulus fibrosus. Most physicians would probably not detect these deep fissures of the annulus. In a rigorous and systematic study of diagnosis based on anesthetic blocks, Schwarzer and colleagues (1995) were able to diagnose over 60% of the LBD cases cited in the study by Bogduk and colleagues (1996) as being internal disc disruption (39%), facet joint pain (15%), or sacroiliac pain (12%). Bogduk and colleagues (1996) stated, "If inappropriate tests such as EMG and imaging are used, nothing will be found in the majority of cases, falsely justifying the impression that nothing can be found." Clearly, we must question the statement that 85% of LBD cases are idiopathic or have no definitive pathoanatomical cause.

At the World Congress for Lumbopelvic Pain in 1995, Professor Bogduk proposed many thought-provoking reasons to caution against defaulting to a psychosocial cause of back troubles, as follows:

- First consult: The usual practice of a 10-minute consult is simply poor practice because it provides no chance for a full assessment of causal mechanisms.
- Evidence-based management: Faith in passive interventions continues, leading to frustration in all parties given poor efficacy.
- Issue of certificate: Rather than engaging patients, uncovering causes of their troubles, and so on, a clinician makes the decree with the issuance of a certificate for time off, light duty, or so forth.
- Workplace intervention: Was the occupationally related cause of the back problem addressed?
- Correct treatment: Was the treatment appropriate or more a consequence of convenience or what may be socially correct?
- Investigations: Prolonged pain requires a thorough investigation. False positives from tests are an iatrogenic nightmare, and their possibility must be considered in these cases.
- Treatment of chronic LBP: Simply the wrong treatment or therapeutic exercise was pursued.
- Lodging a claim: The compensation board has made an incorrect decision.
- Lawyers: Encourage retainment of the disability for maximal claim.
- Expert witness: False witness claiming the cause to be psychosocial.

I must admit that I have had to deal with every one of these points while working with patients, medical management groups, and the legal process. They are important considerations.

Helpful Strategies for Undiagnosed LBDs

Some physicians are clearly frustrated with the delayed improvement of undiagnosed chronic LBD patients. This frustration, together with concern for the financial health of the compensation system, may have motivated the authors of the Fordyce report to emphasize psychosocial modulators rather than organically based variables to explain intolerance to certain types of

activity. But dealing with frustration on the basis of false assumptions will not help the situation. What, then, are more useful approaches to treating undiagnosed chronic cases of LBD?

As noted earlier, many clinicians do not have the expertise or tools to diagnose back troubles at a tissue-based level. Provocative testing will enable many physicians to identify painful motions and loading. By integrating biomechanics with such testing, physicians may be aided in making functional diagnoses.

In a pioneering study, Delitto, Erhard, and Bowling (1995) suggested that appropriately classified back pain sufferers (those with functional classifications rather than tissue-specific diagnoses) do better with specific treatments. Further, as patients progress through the rehabilitation process, they seem to require different treatment approaches. For example, several studies suggest that manipulation can be beneficial for acute short-term troubles, whereas physical therapy and exercise approaches appear better for chronic conditions (Skargren, Carlsson, and Oberg, 1998). Brennan and colleagues (2007) strengthened this impression with a trial showing superior results from matching the therapy and exercise approach with the clinical test results from the patient. For example, those with newer pain, confined to the back, and with stiffer spine segments did better with manipulation (those with longer pain histories did not). Those with repeated episodes, aberrant motion, and painful movement catches did better with stabilization exercise, and those with pain centralization on either flexion or extension did better with exercises matched to the direction of relief. Fritz, Cleland, and Childs (2007) wrote a commentary discussing the evolution of patient classification, which included several refining notes for specific tests. A major remaining limitation of these studies is that none has assessed progressive treatments that change as the patient progresses through the rehabilitation process; rather, they all have assessed only single treatment approaches. Future studies must assess the efficacy of staged programs in which categorized patients follow progressive treatment involving several sequenced approaches.

In summary, the position suggesting that there is no detectable pathoanatomical basis for pain and activity intolerance in some patients and thus that these are functions of only psychosocial variables does not appear to be defensible. Improved tissue-based diagnosis, improved provocative testing and improved functional diagnosis, and a better understanding of the interactions of psychological variables with pathoanatomical variables

support this notion and will continually evolve to help improve treatment outcomes.

Are Biomechanical Variables and Psychosocial Variables Distinct?

Are biomechanical variables and psychosocial variables distinct, or is there an interplay between them that, if understood, would underpin a more evidenced-based intervention program? Most reports have made a clear separation between psychosocial and biomechanical factors. But is there any evidence that psychosocial factors could modulate musculoskeletal loading—or vice versa?

Consider that highly respected pain scientists present volumes of empirical data demonstrating that pain perception is modulated by sensory, neurophysiological, and psychological mechanisms, suggesting that separating the two for analysis is folly (e.g., Melzack and Wall, 1983). Teasell (1997) argued quite convincingly that, although psychological factors have been cited as being causative of pain and disability, in fact, psychological difficulties arise as the consequence of chronic pain (see also Gatchel, Polatin, and Mayer, 1995; Radanov et al., 1994) and disappear upon its resolution (see also Hicks et al., 2005; Mannion et al., 2001; Wallis, Lord, and Bogduk, 1997). Interestingly, when depression is tracked as a variable, some evidence suggests that back pain is a predictor of depression (e.g., Currie and Wang, 2004), yet depression can predispose people to pain (Lepine and Briley, 2004). The evidence is compelling—pain and psychological variables appear to be linked in a bidirectional relationship.

Is there more direct evidence that psychosocial factors are inextricably linked with biomechanical factors? Marras and colleagues (2000) noted that certain personality factors, together with some psychosocial variables, appear to increase spinal loads by up to 27% in some personality types via muscular cocontraction. This appears to occur at moderate levels of loading, whereas biomechanical loading overrides any psychosocial effects under larger task demands. This conclusion was strengthened by a field study linking these general mechanistic observations with reported LBD in workers. In summary, LBDs appear to be associated with both loading and psychosocial factors, and these factors seem to be related and multifactorial. The same conclusion

appears to be valid for many types of chronic pain conditions (Gamsa, 1990).

Does an expectation of impending pain and catastrophizing the expectation influence the eventual pain experience? It appears so. Edwards and colleagues (2008) identified a pathway whereby catastrophizing pain elevated levels of the proinflammatory cytokine interleukin-6. This cytokine increases pain sensitivity and is associated with persisting pain syndromes (Thacker et al., 2007). The next question is how to reduce catastrophizing in patients with back pain? Smeets and colleagues (2006) conducted a randomized controlled trial of 211 patients with back pain with three treatment groups: active physical treatments, cognitive-behavioral treatment, and a combination of the two. All three groups reduced pain and pain catastrophizing, whereas the control group did not. Treating the pain, even with approaches that are physical and do not include specific cognitive-behavioral components, is effective. This series of works provides some insight into the mechanism. This begs one final question: How effective are the purely psychosocial interventions for back pain? Van der Windt and colleagues (2008) reviewed the existing reports of negative efficacy and suggested that a refocus of the research agenda is needed for a better understanding of the potential. It may exist; it may not.

Consider this final tale and logic. My loving dog developed a painful knee. My simply reaching toward her knee caused her to sink her teeth into my arm as she recoiled. Was she displaying psychosocial disturbances causing her pain or a natural response to avoid mechanical loading that she knew would hurt her?

What Is the Significance of First-Time Injury Data for Cause and Prevention?

One of the best indicators of future back troubles is a previous history of back troubles (Bigos et al., 1991; Burton, Tillotson, and Troup, 1989; Troup et al., 1987). This suggests that studies of first-time back trouble episodes may be quite revealing for causative factors. Burton, Tillotson, Symonds, and colleagues (1996) studied police officers in Northern Ireland wearing >8 kg (18 lb) of body armor in a jacket (this additional load was borne by the low back). This group demonstrated a shorter period of time to their first onset of pain when compared to officers in an English force who did not wear the

body armor. The authors also found that spending more than 2 hours per day in a vehicle constituted a separate risk for first-time onset of LBP. A recent study of American soldiers in Afghanistan found that wearing body armor was the most influential predictor of back pain incidence other than having a previous history of back pain (Roy, Lopez, and Piva, 2013). Another survey (Troup, Martin, and Lloyd, 1981) noted that falls among employees across a variety of industries were a common cause of first-time onset and were associated with longer periods of sick leave and a greater propensity for recurrence than were injuries caused by other mechanisms. (Ligamentous damage resulting from this type of loading is discussed in chapter 4.) It is also interesting to note that personal factors appear to play some role in first-time occurrence. Biering-Sorensen (1984) tested 449 men and 479 women for a variety of physical characteristics and showed that those with larger amounts of spine mobility and less lumbar extensor muscle endurance (independent factors) had an increased occurrence of first-time back troubles. Luoto and colleagues (1995) reached similar conclusions. Muscular endurance, and not anthropometric variables, appears to be protective.

Some injuries just happen as the result of motor control errors. (This interesting mechanism is introduced in chapter 4, in which we describe witnessing an injury using videofluoroscopy to view the spine.) These may be considered random events and may be more likely in people with poor motor control systems (Brereton and McGill, 1999).

How Do Biomechanical Factors Affect LBD?

Several approaches have provided evidence of links between biomechanical factors and LBD. A few are summarized in this section.

Mechanical Loading and LBD: Field-Based Risk Factors

Of the epidemiological studies that have focused on kinematic and kinetic biomechanical factors, a few investigated loading of low back anatomical structures. These would be considered to be the strongest evidence. The majority of the studies, however, assessed indirect measures that are linked to spinal loading, such as the presence of static work postures, frequent torso bending and twisting, lifting, pushing or pulling exertions, and exertion repetition. Although tissue overload is the cause of tissue damage and related back troubles, these indirect measures of load merely act as surrogates. The attraction of using surrogate measures rather than direct tissue loads per se for epidemiological study is that they are simpler to quantify and survey in the field. However, trade-offs exist among methodological utility, biological reality, and robustness. Risk factors related to specific tissue-based injury mechanisms are addressed in chapters 4, 6, and 7.

Several issues should be kept in mind when interpreting this literature. Virtually all reviews of the epidemiological literature (e.g., Andersson, 1991; Pope, 1989) have noted that specific job titles and types of work are associated with LBD (although LBD is defined differently in different studies). In particular, jobs characterized by the manual handling of materials, sitting in vibrating vehicles, and remaining sedentary are all linked with LBD. However, this type of data does not reveal much about the links between specific characteristics of the work and the risk of suffering LBD; specifically, a dose-response relationship has not been elucidated. Furthermore, a review of 57 papers that surveyed LBDs revealed no consistency between specific risk factors and the development of those disorders (Ferguson and Marras, 1997). This review demonstrated the large differences in the way surveillance was performed and in risk factor measurements. We are reminded once again that epidemiological approaches

alone will not elucidate the biological pathway of the development of LBDs, a process that must be understood to develop optimal prevention and rehabilitation strategies.

As noted, the majority of risk factors that are addressed in the epidemiological literature (which is surprisingly sparse) are really surrogate factors, or indirect measures, of spine load. These surrogate factors are static work postures; seated work postures; frequent bending and twisting; lifting, pulling, and pushing; vibration (especially seated); and generation of spine power.

- Static work postures. Research has suggested that work characterized by static postures is an LBD risk factor. Although many studies have suggested a link with static work, the key paper on this topic was presented by Punnett and coworkers (1991), who reviewed 1,995 back injury cases from an auto assembly plant. Analyzing jobs for postural and lifting requirements, they found that LBDs were associated with postures that required maintaining mild trunk flexion (defined as the trunk flexed forward from 21 to 45°) (odds ratio = 4.9), postures involving maintaining severe trunk flexion (defined as the trunk being flexed forward greater than 45°) (odds ratio = 5.7), and postures involving trunk twisting or lateral bending greater than 20° (odds ratio = 5.9). Their results suggested that the risk of back injury increased with exposure to these deviated postures and increased with duration of exposure. Deviated postures greatly increase low back tissue loading, particularly when they must be held (Marras et al., 1993; McGill, 1997).
- Seated work postures. Kelsey (1975) linked the seated work posture to a greater risk of LBD. Subsequently, Liira and colleagues (1996) suggested that although white collar (sedentary) workers who must sit for long periods have a greater risk of low back troubles (8% increase in odds risk), active blue collar workers gain some prophylactic effect from sitting down (14% reduction in odds risk). This suggests that variable work, and not too much of any single activity, may reduce mechanically induced low back troubles. Further, nonneutral spine postures while sitting exacerbate the incidence of occupational low back pain, as noted in subway train and tractor operators (Lis et al., 2007) and in dentists who were not able to use backrests (Rafeemanesh et al., 2013). Using backrests can reduce spine loading up to three times by reducing the

muscle activity needed to sit upright (Morl and Bradl, 2013).

- Frequent bending and twisting. A U.S. Department of Labor report (1982) and many more studies (summarized by Andersson, 1981; Marras et al., 1995; Punnett et al., 1991; and Snook, 1982) noted the increased risk of LBD from frequent bending and twisting. In fact, Marras documented in several studies the increased risk of LBD with higher torso velocities (e.g., Marras et al., 1993, 1995). (Note that this is isolated spine motion and not nonspecific torso motion.) Although these studies did not examine a mechanism to explain a link with LBD, the associated motion within the spine will be shown in chapter 4 to form a pathomechanism for very specific disabling LBDs.
- Lifting, pulling, and pushing. In the United States, the National Institute for Occupational Safety and Health report (NIOSH, 1981) provides a good review linking activities requiring lifting, pushing, and pulling with increased risk of LBD.
- Vibration. Vibration, particularly seated vibration, is linked to elevated rates of LBD (e.g., Kelsey, 1975; Lis et al., 2007; Pope, 1989). Our work on cadaveric spines showed that exposure to vibration similar to occupational exposures increased the progression of disc herniation and any resulting herniations (Yates and McGill, 2011).
- Generation of spine power. The concept of power (force $\times$ velocity, or in the context of the spine, spine bending velocity with simultaneous muscle force) has not been well formulated because it is a variable that is calculated rather than measured directly. However, it appears that when the spine muscles are required to generate high forces, the motion needs to be very slow or static to reduce the risk. On the other hand, if the bending velocity is high, then the muscle forces need to be low. Either way, low power is required to minimize risk. This is substantiated by professor Marras' work linking velocity and acceleration to higher risk (Marras, 2008). In addition, in our own laboratory work we have begun experiments loading people's spines isometrically and then with various combinations of load and motion. We had to abandon the experiments as a result of pain generation when higher power levels were reached.

Although all of the risk factors noted here have been epidemiologically linked with an increased incidence of LBD, a subsequent section on tissue

damage provides insight into the mechanisms linking mechanical overload, the onset of pain and disability, and the natural history of these injuries as they pertain to job performance and exposure.

Injury and Lasting Physiological, Biomechanical, and Motor Changes

This section addresses the substantial literature documenting performance deficits and subsequent anatomical changes in LBP populations. The following researchers documented a change in muscular function after injury:

- Hodges and Richardson, 1996, 1999. Delayed onset of specific torso muscles during sudden events may impair the spine's ability to achieve protective stability during situations such as slips and falls. These delays have been further refined to specific subcategories of back pain patients by Silfies and colleagues (2009). Delays in muscular response have been observed to worsen when people focus on their pain, supporting the notion of a neurophysiological pathway (Butler et al., 2010).
- Arendt-Nielsen et al., 1995. Changes in torso agonist–antagonist activity during gait.
- Zedka et al., 1999. Inhibition of back extensors in the presence of pain.
- Freeman, Mascia , and McGill, 2013. Inhibition of the gluteal muscles of the hip in the presence of pain.
- Grabiner , Koh , and Ghazawi , 1992. Asymmetrical muscle output during isokinetic torso extensor efforts that alters spine tissue loading.



Anatomical changes following low back injury include asymmetrical atrophy in the multifidus (Beneck and Kulig, 2012; Hides et al., 1994), particularly at the L4 and L5 segmental levels (Hides et al., 2008), although this is controversial. Fiber changes in the multifidus occur even 5 years after surgery (Rantanen et al., 1993). Further, in a very nice study of 108 patients with histories of chronic LBD ranging from 4 months to 20 years, Sihvonen and colleagues (1997) noted that 50% had disturbed joint motion, and 75% of those with radiating pain had abnormal electromyograms of the medial spine extensor muscles.

Finally, a study of those with a history of low back troubles has shown a wide variety of lingering deficits (McGill et al., 2003). The back troubles were sufficient to cause work loss, but the subjects had been back to work for an average of 270 weeks. Generally, those with a history of troubles were heavier and had disturbances in the flexor–extensor strength ratio and the flexor–extensor endurance ratio together with the lateral bend endurance ratio, had diminished gross flexion range of lumbar motion, and had lingering motor difficulties compromising their ability to balance and bend down to pick up a light object. None of these changes could be considered good.

The broad implication of this work is that a history of low back trouble, even when a substantial amount of time has elapsed since the trouble, is associated with a variety of lingering deficits such that a multidisciplinary intervention approach would be required to diminish their presence. This

collection of evidence is quite powerful in documenting pathomechanical changes associated with chronic LBD. These changes are lasting years—not 6 to 12 weeks!

Optimal Amount of Loading for a Healthy Spine

A lucid interpretation of the data in the epidemiological literature is limited by the fact that the levels at which tissue damage occurs remain obscure. Many are concerned with the known tissue damage that occurs with high magnitudes of load, repetition, and so on. For example, Herrin and colleagues (1986) found that musculoskeletal injuries were twice as likely when workers' lumbar spines were exposed to compressive forces that exceeded 6,800 N (predicted with a biomechanical model). On the other hand, several of the epidemiological studies have not been able to support a link between heavy work, when crudely measured, and the risk of LBD (e.g., Bigos et al., 1986; Porter, 1987). Mitigating factors appear to include the repetition of similar movements and variety in work.

In the discussion of the U-shaped relationship between activity levels and LBD in chapter 1, we saw that stress at optimal levels strengthens the system, whereas too little or too much is detrimental to health. Kelsey (1975) demonstrated a greater-than-expected increase in disc protrusions among sedentary workers. Although many consider sitting a low-load task, in fact, it creates damaging conditions for the disc—the mechanism of which is explained in chapter 4. This has obscured the emergence of much clearer relationships among biomechanical loading, disc herniation, clinical impressions, and the ability to perform demanding work. Further, Videman, Nurminen, and Troup (1990) studied a cross section of retired workers by comparing their LBD history with their magnetic resonance imaging (MRI) scans. A history of back pain and the visible parameters of spinal pathology were least prevalent in workers whose jobs had included moderate activity and most prevalent in workers with either sedentary or heavy work.

Many studies compare just two levels of work—for example, light versus heavy, light versus moderate, or moderate versus heavy. Because any relationship other than a straight line requires more than two points (or levels of activity or levels of loading), such studies cannot illustrate the U-shaped relationship. Consider the data compiled in a thorough epidemiological study

reported by Liira and colleagues (1996) of LBD prevalence and physical work exposures. Even though the authors considered only the upper levels of excessive loading, they concluded that one-quarter of excess back pain morbidity could be explained by physical work exposures. That this is an underestimate is a very real possibility.

Furthermore, the nature of LBD appears to be affected by the type of work. Videman and colleagues (1990) noted a tendency among those who had had sedentary careers to have marked disc degeneration in later years, whereas those who had performed heavy work (defined as not only lifting but also requiring large trunk motions) tended to have classic arthritic changes in the spine (stenosis, osteophytosis, etc.). Recent work by the same group (Wang et al., 2012) noted a strong link between vertebral end-plate damage and pain frequency, secondary disc degeneration, and Schmorl's nodes in those with occupations classified as heavier (meaning more demanding). In a similar study, Battie and colleagues (1995) reported an apparent contribution of genetic factors to various age-related changes in the spine using monozygotic twins, given significantly greater similarities in spinal changes than would be expected by chance.

Porter (1987, 1992) performed two studies that furthered the notion of an optimal loading level for health. The first study tracked miners and nonminers treated at hospitals for back pain. Although significantly more miners reported back trouble compared with nonminers, significantly fewer were diagnosed with disc protrusions, whereas significantly more were reported to have stenosis and nerve root entrapment (conditions associated more with the arthritic spine according to the data of Videman, Nurminen, and Troup, 1990). The second study evaluated questionnaire results from 196 patients with symptomatic disc protrusion and 53 with root entrapment syndrome. They were asked about their history of heavy work between the ages of 15 and 20. Significantly more subjects with disc protrusion had done no heavy physical work in those early years. By contrast, more of those with nerve entrapment syndrome had done 5 full years of heavy work between 15 and 20 years of age. These collective data suggest that different work demands cause different spine conditions and perhaps that the optimal loading is different for different tissues. Nonetheless, the optimal activity appears to be varied work at a moderate level between sedentary and heavy work.

What constitutes an optimal load—a load that is not too much, not too

little, not too repetitive, and not too prolonged? Currently, we are in need of assessment tools for determining optimal load to guide the intervention toward optimal health.

Links Between Personal Factors and LBD

Some personal factors such as muscle endurance (not strength) and less spine range of motion (not more) are prophylactic for future back troubles. These are addressed in part III, Low Back Rehabilitation. A few other personal factors are noted here.

A few personal factors appear to affect spine tissue tolerance according to the existing literature; age and gender are two examples. Jager, Luttmann, and Laurig (1991) compiled the available literature on the tolerance of lumbar motion units to bear compressive load that passed their inclusion criteria. Their results revealed that when males and females were matched for age, females were able to sustain only approximately two-thirds of the compressive loads of males. Furthermore, Jager and colleagues' data showed that within a given gender, the 60-year-old spine was able to tolerate only about two-thirds of the load tolerated by the 20-year-old spine. Keep in mind that age and gender are very simple factors.

It appears that other personal factors, such as poor motor control system fitness, can lead to a back injury during ordinarily benign tasks such as picking up a pencil from the floor. The modeling data of Cholewicki and McGill (1996) suggest that the spine can easily buckle during such a task. When the muscle forces are inherently low, a small motor error can cause rotation of a single spinal joint, placing all bending moment support responsibility on the passive tissues. Such scenarios do not constitute excessive tasks, but patients often report them to clinicians as the event that caused their injury. This phenomenon will not be found in the scientific literature, however. Many medical personnel would not record this event as the cause of injury because in many jurisdictions it would not be deemed a compensable injury. These types of injuries seem to be more influenced by the fitness of a person's motor control system than by factors such as strength. McGill, Sharratt, and Seguin (1995) noted that people differ in their ability to hold a load in their hands and breathe heavily. This is very significant because the muscles required to be continuously active to support the spine (and prevent buckling) are also used to breathe by rhythmically contracting. Those who must use their muscles to breathe in this way sacrifice spine stability. Interestingly, we have measured changing strategies in those with compromised lung elasticity (e.g., smokers, those with

emphysema) who use their back extensors to assist with lung inflation, resulting in the perverse effect of enhancing stability at the expense of wearing down the endurance capacity of their back muscles (Wang and McGill, 2008). All of these deficient motor control mechanisms heighten biomechanical susceptibility to injury or reinjury (Cholewicki and McGill, 1996) and are highly variable personal characteristics.

As noted previously, muscle endurance is predictive of future episodes of back pain, however those studies measured back extensor endurance. A collection of works measuring gluteal and lateral torso muscles suggests that endurance deficits predict those who will develop pain (Gallagher, Nelson-Wong, and Callaghan, 2011; Marshall, Patel, and Callaghan, 2011; Nelson-Wong and Callaghan, 2010).

Additional factors other than simple load magnitude appear to modulate the risk of tissue damage. The mechanism of disc herniation provides an example. Although disc herniations have been produced under controlled conditions (e.g., Gordon et al., 1991), they have not been produced consistently. Our lab has been able to consistently produce disc herniations by mimicking spine motion and load patterns seen in workers (Callaghan and McGill, 2001). Specifically, it appears that only a very modest amount of spine compression force is required (only 800-1,000 N), but the spine specimen must be repeatedly flexed—mimicking repeated torso–spine flexion from continual bending to a fully flexed posture. The main relevance for this issue is that the way workers elect to move and bend influences the risk of disc herniation. This highlights the need to examine how workers move in standardized tests at the beginning of their careers while they still have virgin backs, making it possible to determine cause and effect. Recent evidence suggests that those with a history of back troubles are more likely to lift flexing the spine and not the hips, increasing the risk of back damage (McGill et al., 2003).

Studies of Exercise Intervention

So far we have introduced the notions that exercise alone is not optimal, and that effort must be directed to eliminate whatever is causing the symptoms. Also, a form of exercise that helps one person will hurt another. Each person is unique in terms of what causes pain, and the dosage that will cause less pain. We have reviewed how exercise can wind down pain sensitivity, strengthen injured tissue, enhance function, and improve movement to reduce the risk of tissue irritation—but only when the type and dose of exercise is tuned to the individual. Therefore, studies of exercise trials in which each person is given the same exercise and dose should not be expected to yield positive results, yet they do. In fact, the exercise studies have the strongest support for positive patient outcome (Bigos et al., 2009), although the effectiveness is for chronic, not acute, pain (Van Middlekoop et al., 2010). Some exercise trials used machines to isolate spine and back muscle movement and claimed good results in reducing pain. However, when these exercises were compared to nonmachine exercises, they were not shown to be superior (Helmhout et al., 2008). Concerns and issues with using machines are introduced later. Again, the studies usually do not report adequate descriptions of the actual exercises, dosage, or patient matching. The studies control the training sessions per week and do not fine-tune exercises for individual pain patterns. When exercise programs are individualized for injury and pain history, current fitness levels or deficits, training goals and other personal descriptors, efficacy increases.

Suni and colleagues implemented an exercise program combined with counseling and coaching to remove poor movement (documented in this book); effectiveness was shown for middle-aged men (2006) and military personnel (2013). Performing the big three exercises, documented in this book, reduced injury rates in women collegiate gymnasts (Durall et al., 2009). Note: Big three exercises was a term coined to describe the original stabilization exercises that we showed to be superior in challenging the spine stabilizing musculature in a way that spared the spine of applied load (McGill, 1998).

Practical Application: What the Evidence Supports

In summary, it is interesting to consider why only some workers become patients. There is no question that damage to tissue can be caused by excessive loading, and that damage causes pain. However, pain is a perception that is modulated by psychosocial variables in addition to physiological injury. Clearly, both psychosocial and biomechanical variables are associated with LBD and are important in preventing low back injury and the ensuing chronicity; collectively, the evidence from several scientific perspectives is overwhelming. The relative importance of either is often difficult to compare across studies because the metrics for each are different—biomechanical variables are reported in newtons, newton-meters, numbers of cycles, and so forth, whereas psychosocial variables are reported in ordinal scales linked to perception (independent risk factors can be compared using odds ratios). Some influential reports have ignored biomechanical evidence and promoted psychosocial variables as more important. However, no study of psychosocial variables has been able to conclusively establish causal links—only association. Some biomechanically based studies, together with the chronic pain literature, are strongly convincing in their establishment of both association and causality. Thank goodness! We can now proceed.

Back injury can begin with damage to one tissue, but this changes the biomechanical function of the joint. Tissue stresses change and other tissues become involved, leading to progressive deterioration with time. Tissue damage does not always result from too high a load magnitude. In the case of disc herniation, repetitive motion, even in the absence of large loads, seems to be a significant causative mechanism. The notion that tissues heal within 6 to 12 weeks and that longer-lasting work intolerance has no pathoanatomical basis appears to be false; tissue injury data and the science of chronic pain mechanisms strongly suggest otherwise.

Understanding the role of biomechanical, psychosocial, and personal factors as well as their interrelationships will build the foundation for better prevention in the future. The challenge is to develop variable tolerance guidelines, psychosocial guidelines, and higher-level medical practice codes.

On balance, the evidence supports the following statements:

- Biomechanical factors are linked to both the incidence of first-time low

back troubles and absenteeism, and subsequent episodes.

- Psychosocial factors appear to be important as well, but may be more related to episodes after the initial back-related episode.
- Psychosocial and biomechanical factors appear to influence each other, in terms of both causation of work absence and the course of recovery.
- Treatment to reduce back pain often leads to a reduction in psychosocial issues, not the other way around. There is overwhelming evidence that once patients can obtain a sound night's sleep and can eliminate low-level chronic pain, their psychological constitution and mental toughness return.
- The relationship between loading and LBDs appears to be U-shaped, and the optimal load is at a moderate level.
- Low back tissue damage can initiate a cascade of changes that may cause pain and intolerance to certain activities, and these changes may be disruptive for up to 10 years in an unfortunate few.
- Many types of tissue damage can escape detection in vivo. Even gross damage visible during dissection is often not visible on medical images. Thus, nonspecific diagnosis does not rule out the presence of mechanical damage and must not be used to imply that mechanical factors are not related to LBD.

What Works Clinically

The approach developed in this book is what I use daily with patients and in my consulting for others. We follow up with every patient I see, and we know our clinical score. The patients who are referred to me must qualify by having pain and having been seen by several clinicians who either failed to get results or made the situation worse. These patients are considered the most difficult. I enjoy the challenge.

These challenging back patients have only one thing in common: a track record of failing to get better. Based on their intolerance (the motion or load that triggers symptoms), our follow-up revealed excellent outcome (i.e., cured with no remaining issue) as follows:

- 48% of patients with flexion intolerance
- 75% of patients with extension intolerance
- 80% of patients with flexion and extension intolerance
- 33% of patients with flexion and compression intolerance

When categorizing these people based on their diagnosis (the diagnosis resulted from our series of tests), our follow-up revealed excellent outcome (cured with no remaining issue) as follows:

- 47% of discogenic patients
- 78% of disc and facet patients

Responses to the question Did you experience an improvement after seeing McGill? were as follows:

- 19%: No improvement
- 23%: Moderate improvement
- 58%: Substantial improvement

Responses to the question Did you get worse? were as follows:

- 9%: Yes
- 91%: No

Responses to the question Do you continue with the exercise prescription? were as follows:

- 71%: Yes
- 21%: No (8% didn't answer)

What may be surprising is that I saw each patient for 3 hours, only once. The sessions included an assessment of movement quality and pain provokers, followed by education and instruction on moving to avoid their specific pain trigger, and a session with notes outlining a graded and progressive therapeutic exercise program tuned for them.

Others have implemented the approach in this book on a wide scale. You should expect better results with patients who have not yet become the clinical “failures.” Suni and colleagues (2013) documented the outcome of a 6-month neuromuscular exercise and counseling program for new conscripts in the Finnish military who entered with no back pain. Days lost due to back pain were significantly reduced in the experimental subjects compared to the controls.

A Final Note

After warning you not to skip this traditionally boring chapter on epidemiological approaches, I hope that you have found value. Understanding the foundation for the opinions you may hear about back pain makes you an educated consumer. Scientists will find what they look for, and groups will ignore specific evidence when it suits their agendas. I have always found that people make better decisions when informed with fair and balanced evidence.